



Department of Electrical and Electronics Engineering

ELE 214: Electronics Laboratory - I
Multistage Amplifier Design and Fabrication

Term Project - Spring 2026

Student: Baturay Kafkas
Student ID: 2240357068

Contents

1	Abstract	3
2	Objective	3
2.1	Buffer Stage	3
2.2	Equalizer Stage	3
2.3	Gain Stage	3
3	Research and Analytical Calculations	3
3.1	Definitions	3
3.2	Transistor Choice	4
3.3	Circuit Design	7
3.3.1	Buffer Stage	7
3.3.2	Equalizer Stage	8
3.3.3	Gain Stage	8
3.4	Pros and Cons of Design	8
4	Simulation	9
4.1	Buffer Stage Simulation	9
4.2	Equalizer Stage Simulation	10
4.3	Gain Stage Simulation	12
4.4	Multistage Simulation	13
5	Fabrication	13
5.1	Breadboard Testing	14
5.2	Assembly on Pertinax	17
6	Conclusion	19
7	References	19
A	Handwritten Calculations and Schematics	20
B	Figures and Tables	21

1 Abstract

This project demonstrates the knowledge gathered in ELE230 - Electronics I, ELE214 - Electronics Laboratory I, and ELE226 - Circuit Theory II in the 2025-2026 spring academic term. It allows the student to gain hands-on experience with electrical components, acquire skills, and put theory into practice. The lists of figures and tables are given in Appendix B.

2 Objective

The goal of this project is to be able to use the two types of transistors, BJT and MOSFET, and other electrical components to form up an audio amplifier with a specific gain. In this project, the use of operational amplifier is to be avoided. The system must include a buffer stage, an adjustable band-pass equalizer, and a gain stage.

As instructed, at least one BJT and one MOSFET should be used. In this project, the BJT is going to be used for buffer stage and MOSFET for the gain stage.

2.1 Buffer Stage

Since this stage has unity gain, an emitter follower configuration may be used. It has a large input resistance and low output resistance [1].

2.2 Equalizer Stage

This stage is to be designed using two capacitors. An RLC circuit could also do the same job. Since a capacitor can be easily found on the market, it is preferred to utilize capacitors instead.

2.3 Gain Stage

Our aim is to increase the output voltage. Therefore, a common-source configuration can be used. In this stage, the audio is ensured to be amplified. Since no speaker is going to be used, we can directly measure the output voltage in this stage.

3 Research and Analytical Calculations

3.1 Definitions

- **Amplifier:** An electronic device that is used to increase the magnitude of an input signal.
- **Buffer stage:** The stage that transfers a signal from one stage to another without altering its voltage level (i.e., it has unity gain) while preventing interaction between the stages.
- **Equalizer stage:** A circuit or processor that attenuates or boosts specific frequencies. It determines the frequency range that will be conveyed to the other stage.
- **Gain stage:** The stage where the signal level is increased to a desired value.
- **Multistage amplifier:** An amplifier formed by connecting two or more amplifier stages in series. The stages are said to be in a cascade connection.

- **Role of Capacitors:** In an audio amplifier, capacitors are used mainly in two roles: coupling capacitors and bypassing capacitors.

Coupling capacitors pass AC signals while blocking DC between stages and prevent bias disturbance.

Bypass capacitors stabilize gain by providing a low-impedance path to ground for AC, disabling the degeneration resistor.

- **MOSFET vs BJT Differences:** Table 3.1 and Figure 3.1 show the differences.

Table 3.1: Comparison between BJT and MOSFET

Feature	BJT	MOSFET
Classification	NPN, PNP	NMOS, PMOS
Control Type	Current-controlled device	Voltage-controlled device
Terminals	Emitter, base, collector	Source, gate, drain, body
Input Impedance	Low ($k\Omega$ range)	Very high ($M\Omega$ to $G\Omega$)
Input Current	Continuous base current	Almost zero gate current
Regions	Active, saturation, cutoff	Linear (ohmic, triode), saturation
Power Consumption	Higher	Lower
Thermal Stability	Poor	Better
Noise Performance	Higher noise	Lower noise
Typical Applications	Analog amplifiers	Digital circuits, power electronics

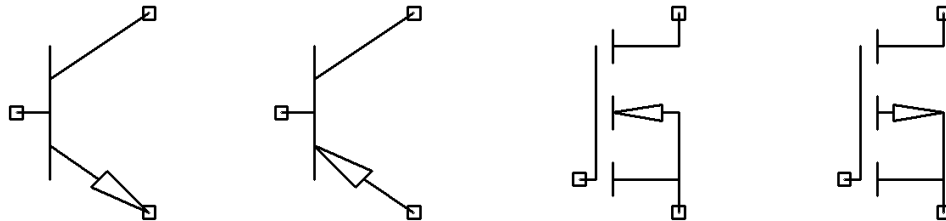


Figure 3.1: LTspice Symbols From Left to Right: NPN, PNP, NMOS, PMOS

3.2 Transistor Choice

In this design, we decided to use the 2N2222 and 2N7000 models. They are widely available, low-cost, and well-documented components that are suitable for an electronics project that a student may design. The BJT provides a reliable solution for small-signal amplification. As for the MOSFET, it offers efficient switching with minimal drive requirements and enhances the output voltage.

Table 3.2 [2], Figure 3.2, and Figure 3.3 show the properties of the 2N2222/2N2222A transistor. Table 3.3 [3] shows the properties of the 2N7000 transistor.

Table 3.2: Electrical Characteristics of 2N2222/2N2222A NPN Transistor ($T_a = 25^\circ\text{C}$ unless otherwise noted)

Symbol	Parameter	Conditions	Min	Max	Unit
OFF CHARACTERISTICS					
$V_{(BR)CBO}$	Collector-Base Breakdown Voltage	$I_C = 10\ \mu\text{A}$ (2N2222)	60	–	V
		$I_C = 10\ \mu\text{A}$ (2N2222A)	75	–	V
$V_{(BR)CEO}$	Collector-Emitter Breakdown Voltage	$I_C = 10\ \text{mA}$ (2N2222)	30	–	V
		$I_C = 10\ \text{mA}$ (2N2222A)	40	–	V
$V_{(BR)EBO}$	Emitter-Base Breakdown Voltage	$I_E = 10\ \mu\text{A}$ (2N2222)	5	–	V
		$I_E = 10\ \mu\text{A}$ (2N2222A)	6	–	V
I_{CBO}	Collector-Base Cutoff Current	$V_{CB} = 50\ \text{V}$ (2N2222)	–	10	nA
		$V_{CB} = 60\ \text{V}$ (2N2222A)	–	10	nA
ON CHARACTERISTICS					
h_{FE}	DC Current Gain	$V_{CE} = 10\ \text{V}, I_C = 0.1\ \text{mA}$	35	–	–
		$V_{CE} = 10\ \text{V}, I_C = 1\ \text{mA}$	50	–	–
		$V_{CE} = 10\ \text{V}, I_C = 10\ \text{mA}$	75	–	–
		$V_{CE} = 10\ \text{V}, I_C = 150\ \text{mA}$	100	300	–
		$V_{CE} = 10\ \text{V}, I_C = 500\ \text{mA}$ (2N2222)	30	–	–
		$V_{CE} = 10\ \text{V}, I_C = 500\ \text{mA}$ (2N2222A)	40	–	–
$V_{CE(sat)}$	Collector-Emitter Saturation Voltage	$I_C = 150\ \text{mA}, I_B = 15\ \text{mA}$ (2N2222)	–	0.4	V
		$I_C = 150\ \text{mA}, I_B = 15\ \text{mA}$ (2N2222A)	–	0.3	V
		$I_C = 500\ \text{mA}, I_B = 50\ \text{mA}$ (2N2222)	–	1.6	V
		$I_C = 500\ \text{mA}, I_B = 50\ \text{mA}$ (2N2222A)	–	1.0	V
$V_{BE(sat)}$	Base-Emitter Saturation Voltage	$I_C = 150\ \text{mA}, I_B = 15\ \text{mA}$ (2N2222)	–	1.3	V
		$I_C = 150\ \text{mA}, I_B = 15\ \text{mA}$ (2N2222A)	0.6	1.2	V
		$I_C = 500\ \text{mA}, I_B = 50\ \text{mA}$ (2N2222)	–	2.6	V
		$I_C = 500\ \text{mA}, I_B = 50\ \text{mA}$ (2N2222A)	–	2.0	V
DYNAMIC CHARACTERISTICS					
f_T	Gain Bandwidth Product	$I_C = 20\ \text{mA}, V_{CE} = 20\ \text{V}, f = 100\ \text{MHz}$	250	–	MHz
C_{ob}	Collector Output Capacitance	$V_{CB} = 10\ \text{V}, f = 1\ \text{MHz}$	–	8	pF

Table 3.3: Electrical Characteristics of 2N7000 MOSFET ($T_C = 25^\circ C$ unless otherwise noted)

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
OFF CHARACTERISTICS						
BV_{DSS}	Drain-Source Breakdown Voltage	$V_{GS} = 0\text{ V}, I_D = 10\text{ }\mu\text{A}$	60	–	–	V
I_{DSS}	Zero Gate Voltage Drain Current	$V_{DS} = 48\text{ V}, V_{GS} = 0\text{ V}$	–	–	1	μA
		$V_{DS} = 48\text{ V}, V_{GS} = 0\text{ V}, T_C = 125^\circ C$	–	–	1	mA
I_{GSSF}	Gate-Body Leakage, Forward	$V_{GS} = 15\text{ V}, V_{DS} = 0\text{ V}$	–	–	10	nA
I_{GSSR}	Gate-Body Leakage, Reverse	$V_{GS} = -15\text{ V}, V_{DS} = 0\text{ V}$	–	–	-10	nA
ON CHARACTERISTICS						
$V_{GS(th)}$	Gate Threshold Voltage	$V_{DS} = V_{GS}, I_D = 1\text{ mA}$	0.8	2.1	3	V
$R_{DS(on)}$	Static Drain-Source On-Resistance	$V_{GS} = 10\text{ V}, I_D = 500\text{ mA}$	–	1.2	5	Ω
		$V_{GS} = 10\text{ V}, I_D = 500\text{ mA}, T_C = 125^\circ C$	–	1.9	9	Ω
		$V_{GS} = 4.5\text{ V}, I_D = 75\text{ mA}$	–	1.8	5.3	Ω
$V_{DS(on)}$	Drain-Source On-Voltage	$V_{GS} = 10\text{ V}, I_D = 500\text{ mA}$	–	0.6	2.5	V
		$V_{GS} = 4.5\text{ V}, I_D = 75\text{ mA}$	–	0.14	0.4	V
g_{FS}	Forward Transconductance	$V_{DS} = 10\text{ V}, I_D = 200\text{ mA}$	100	320	–	mS
DYNAMIC CHARACTERISTICS						
C_{iss}	Input Capacitance	$V_{DS} = 25\text{ V}, V_{GS} = 0\text{ V},$	–	20	50	pF
C_{oss}	Output Capacitance	$f = 1.0\text{ MHz}$	–	11	25	pF
C_{rss}	Reverse Transfer Capacitance		–	4	5	pF
t_{on}	Turn-On Time	$V_{DD} = 15\text{ V}, R_L = 25\text{ }\Omega,$ $I_D = 500\text{ mA}, V_{GS} = 10\text{ V},$ $R_{GEN} = 25\text{ }\Omega$	–	–	10	ns
t_{off}	Turn-Off Time	$V_{DD} = 15\text{ V}, R_L = 25\text{ }\Omega,$ $I_D = 500\text{ mA}, V_{GS} = 10\text{ V},$ $R_{GEN} = 25\text{ }\Omega$	–	–	10	ns

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

1. Pulse Test: Pulse Width = 300 μs , Duty Cycle $\leq 2\%$.

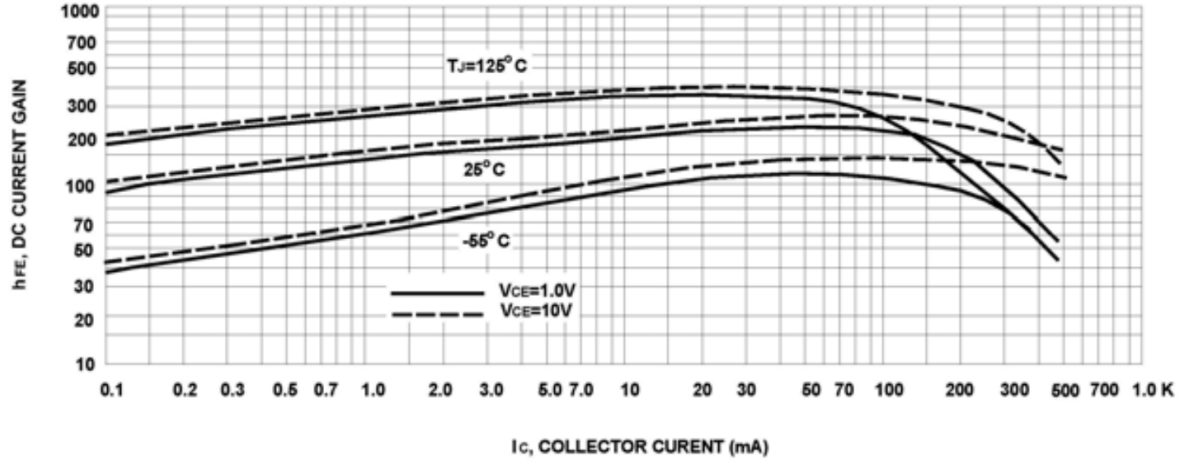


Figure 3.2: Beta Values for Different Temperature Values

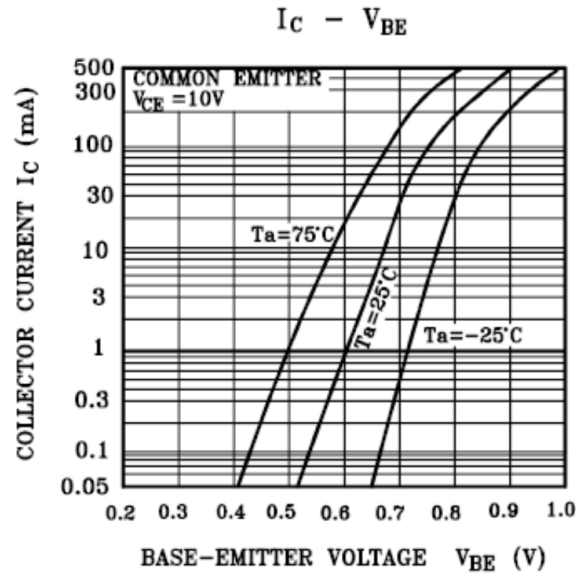


Figure 3.3: Relationship between the Collector Current and the Base-Emitter Voltage

3.3 Circuit Design

Summary of design parameters based on student ID (ending in 68):

- $f_{low} = 680 \text{ Hz}$ and $f_{high} = 13200 \text{ Hz}$
- Target Gain: 8

Calculations for each stage and schematics are given in Appendix A.

3.3.1 Buffer Stage

The following assumptions are made for this stage:

$$I_C = 10 \text{ mA}, \quad \beta = 200, \quad V_{BE, ON} = 0.7 \text{ V}, \quad I_{R1} \approx I_{R2} \geq 10 \text{ mA}.$$

The β value is taken as 200 by imposing the assumption $I_C = 10$ mA for $T = 25^\circ\text{C}$ from Figure 3.2. By imposing the same assumption, the V_{BE} value is taken as approximately 0.7 V from Figure 3.3. Table 3.4 lists the components for this stage.

Table 3.4: Buffer Stage Component Values

Component	Value	Description
R_1	6.8 k Ω	Base bias resistor (connected to DC)
R_2	10 k Ω	Base bias resistor (grounded)
R_3	470 Ω	Emitter resistor
C_1	1 μF	Input coupling capacitor
C_2	10 μF	Output coupling capacitor
Q_1	2N2222	NPN BJT transistor

3.3.2 Equalizer Stage

Table 3.5 lists the components for this stage.

Table 3.5: Equalizer Stage Component Values

Component	Value	Description
R_4	1.2 k Ω	Low-pass resistor
C_4	10 nF	Low-pass capacitor
R_5	24 k Ω	High-pass resistor
C_5	10 nF	High-pass capacitor

3.3.3 Gain Stage

Like the buffer stage, the following assumptions are made:

$$K = 0.085 \text{ A/V}^2 \quad V_{DS} = 2 \text{ V}, \quad V_{TH} = 1.6 \text{ mA}, \quad I_{D, \text{SAT}} = 2 \text{ mA}$$

Table 3.6 lists the components for this stage.

Table 3.6: Gain Stage Component Values

Component	Value	Description
R_6	560 k Ω	Gate bias resistor (connected to DC)
R_7	180 k Ω	Gate bias resistor (grounded)
R_8	2 k Ω	Collector resistor
R_9	220 Ω	Source resistor
C_6	1 μF	Input coupling capacitor
C_7	10 μF	Output coupling capacitor
Q_2	2N7000	N-channel MOSFET transistor

3.4 Pros and Cons of Design

BJTs provide relatively high transconductance and good voltage gain at low currents. However, in this circuit, the BJT is used as a buffer based on preferences. The unbypassed emitter resistor improves thermal stability and linearity by introducing local negative feedback into the circuit.

The passive filters are highly predictable, stable, and very linear because there are no active semiconductor nonlinearities involved. They can be useful for AC coupling between stages or shaping the overall response of the circuit. The tradeoff is that passive networks always introduce attenuation, so some signal strength is lost.

One advantage of the MOSFET is the extremely high input impedance of the gate. Because there is essentially no gate current flowing, the input signal experiences little load. However, the gain may become less than the theoretical value compared to the one obtained with a BJT.

4 Simulation

The model 2N7000 didn't exist in the current LTspice environment. Therefore, a new spice directive [4] is used. The directive for this model:

```
.model 2N7000 VDMOS (Rg=3 Vto=1.6 Rd=0 Rs=.75 Rb=.14 Kp=.17
mtriode=1.25 Cgdmax=80p Cgdmin=12p Cgs=50p Cjo=50p Is=.04p
mfg=Fairchild Vds=60 Ron=2 Qg=1.5n)
```

In each stage simulation, the directive `.tran 0 2m 0 0.01m` is used and an input voltage of 20 mV and the source frequency of 3 kHz, the approximate frequency transmitted the most, are assumed. At the buffer and gain stages, a measuring instrument with 1 M Ω internal resistance is assumed to be used to prevent the DC offset despite being not shown in the figures.

4.1 Buffer Stage Simulation

The stage design is given in Figure 4.1 and its simulation is given in Figure 4.2.

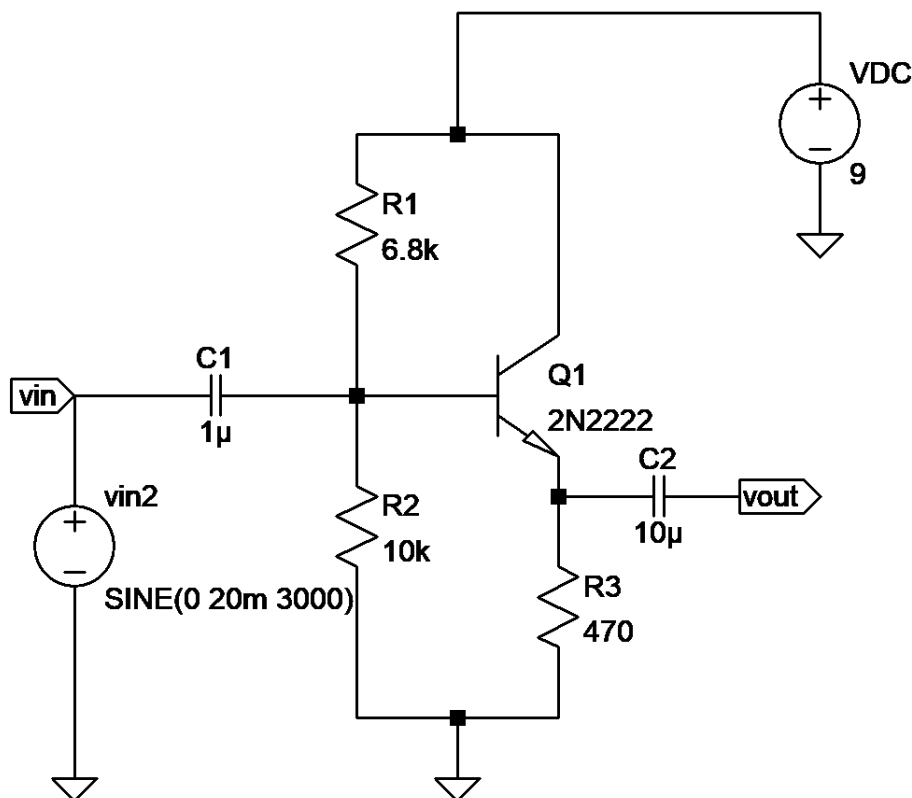


Figure 4.1: Buffer Stage Simulation Circuit

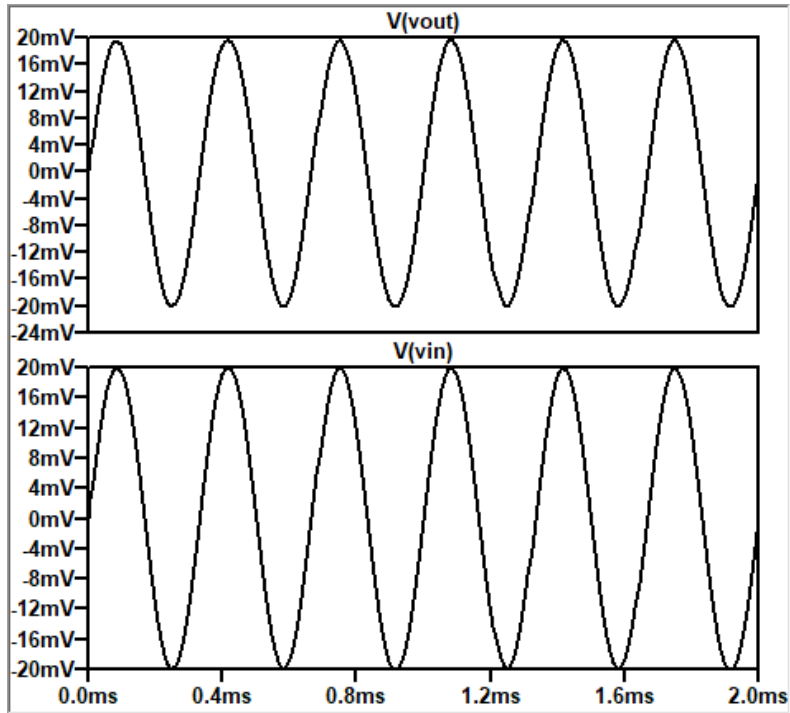


Figure 4.2: Buffer Stage Simulation

As expected, the voltage gain is $A_v = 1 \text{ V/V}$.

4.2 Equalizer Stage Simulation

The stage design is given in Figure 4.3.

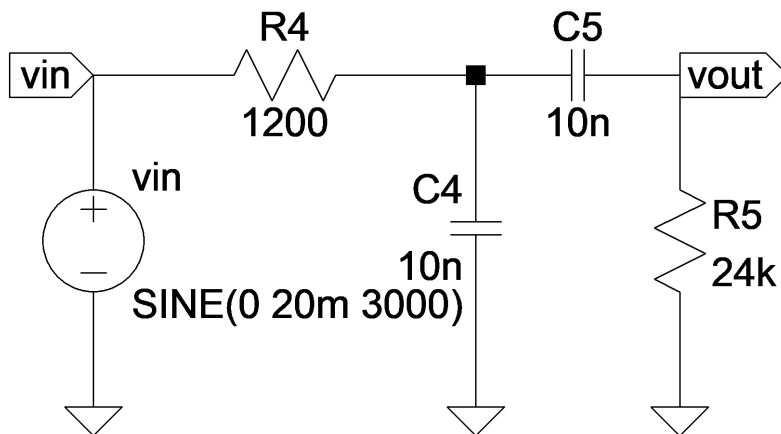


Figure 4.3: Equalizer Stage Simulation Circuit

The input and output voltage are simulated in Figure 4.4 and the amplitude of the output voltage is given in Figure 4.5.

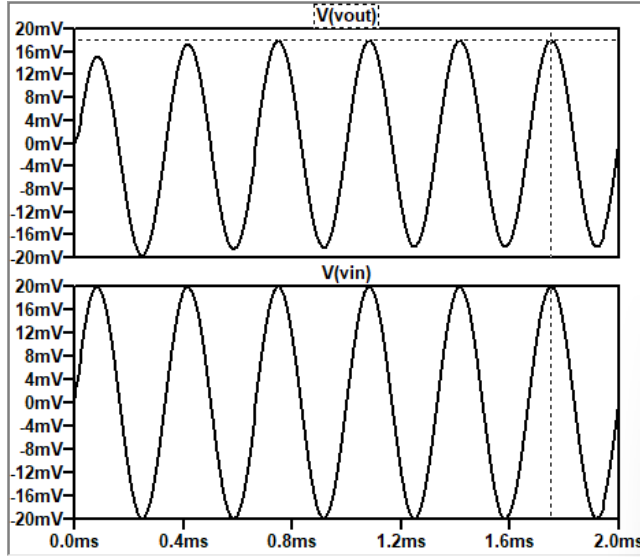


Figure 4.4: Equalizer Stage Simulation (Voltage Characteristics)

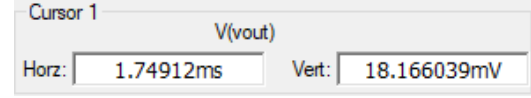


Figure 4.5: Equalizer Stage Simulation (Voltage Characteristics) Comparison

For $f = 3$ kHz and $V_{in} = 20$ mV, the input voltage is attenuated by approximately 9.8%. Therefore, we may infer that at least almost ten percent of the signal is attenuated for all frequency values.

The Bode plot is given in Figure 4.6 and the cutoff frequencies are given in Figure 4.7.

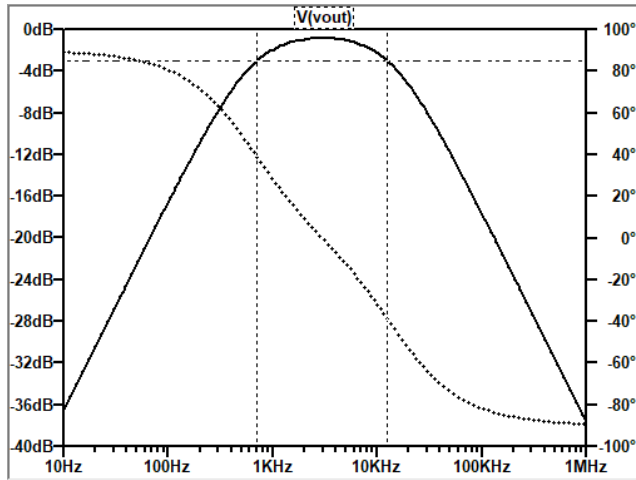


Figure 4.6: Equalizer Stage Simulation (Bode Plot)

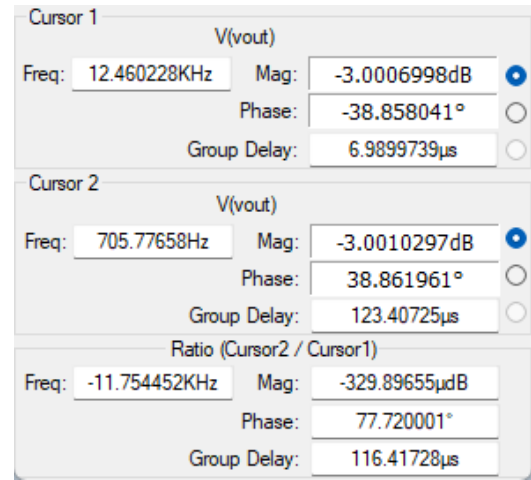


Figure 4.7: Equalizer Stage Simulation (Bode Plot) Result

The cutoff frequencies are found to be $f_{lower} \approx 706$ Hz and $f_{upper} \approx 12.460$ kHz.

4.3 Gain Stage Simulation

The stage design is given in Figure 4.8 and its simulation is given in Figure 4.9. The result is given in Figure 4.13.

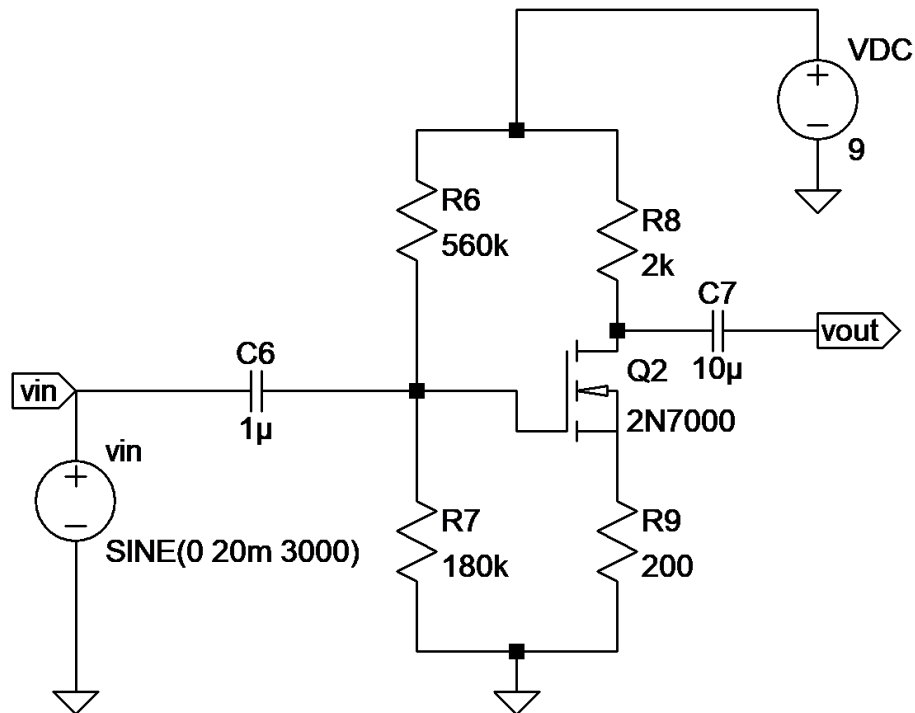


Figure 4.8: Gain Stage Simulation Circuit

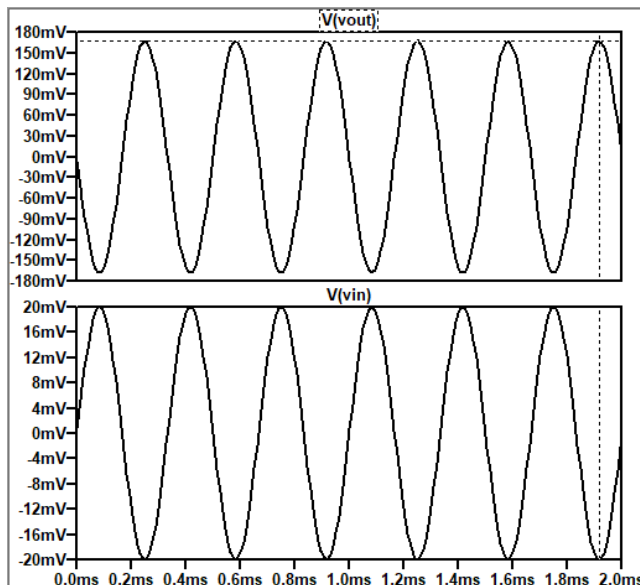


Figure 4.9: Gain Stage Simulation

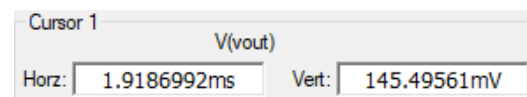


Figure 4.10: Gain Stage Simulation Result

Thus the voltage gain is found to be $A_v \approx -8.38\ \text{V/V}$. As expected, there is a 180° phase shift. The gain obtained from the simulation differs by a small amount from the desired value.

4.4 Multistage Simulation

The design is given in Figure 4.11 and its simulation is given in Figure 4.12. The result is given in Figure 4.10.

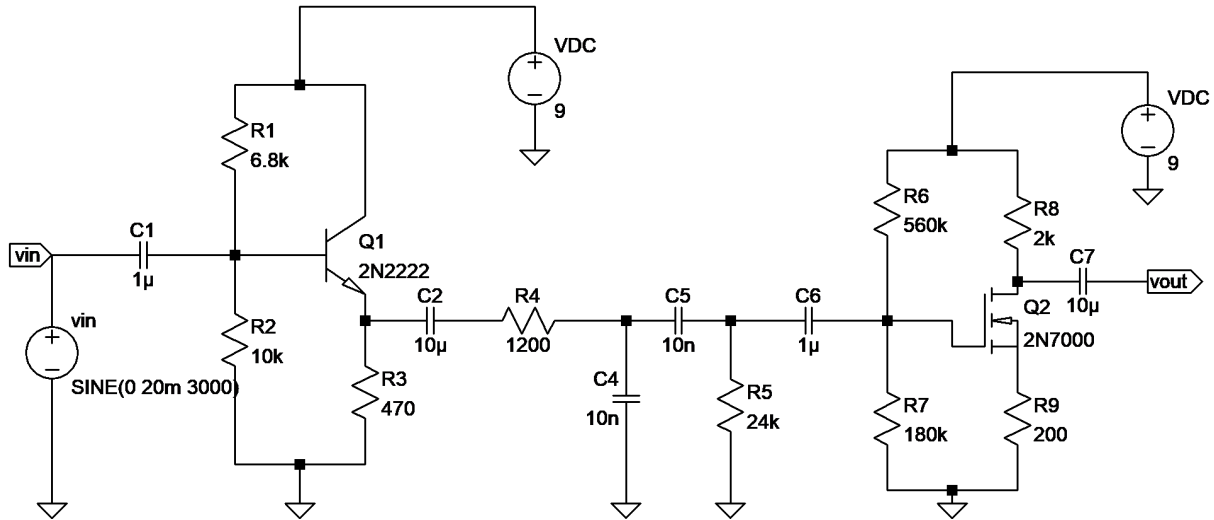


Figure 4.11: Multistage Simulation Circuit

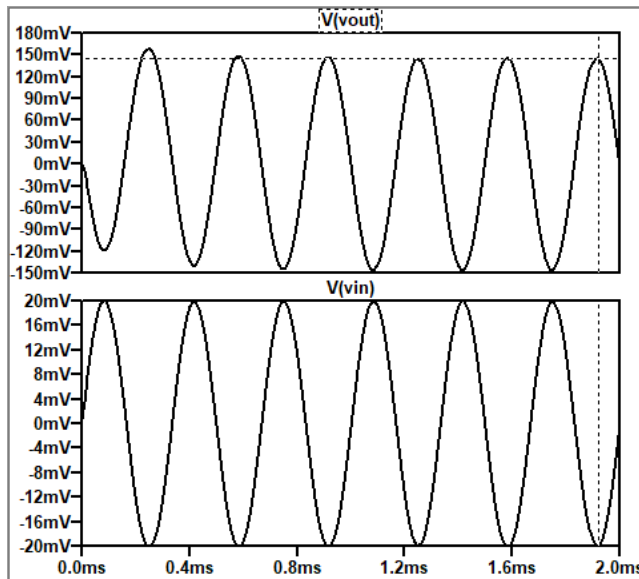


Figure 4.12: Multistage Simulation

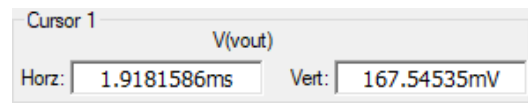


Figure 4.13: Multistage Simulation Result

Thus the overall voltage gain is found to be $A_v \approx -7.27 \text{ V/V}$. As expected, there is a 180° phase shift. The overall voltage gain obtained from the simulation differs by 1.11 from the gain stage output gain and 0.73 from the desired value in magnitude.

5 Fabrication

The circuit components were purchased from the shopkeepers on Konya Street. Since it would take some time to form up a brand new PCB, we decided to use a pertinax to solder and assemble the components. Before that, the circuit is tested on a breadboard to ensure that it is operating as expected. Timestamps of photos are given in Appendix B.

5.1 Breadboard Testing

Figure 5.1 shows the assembly of the buffer stage on the breadboard and Figure 5.2 demonstrates whether the buffer stage works properly.

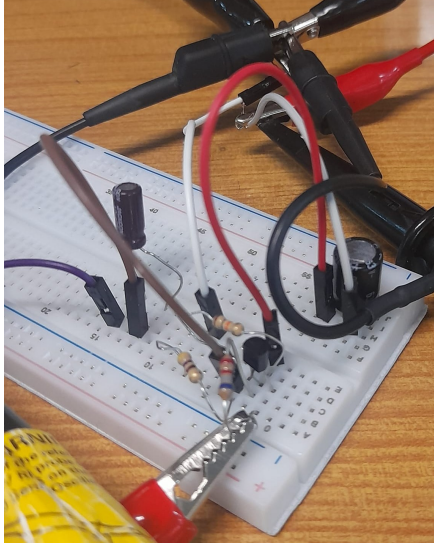


Figure 5.1: Buffer Stage on the Breadboard

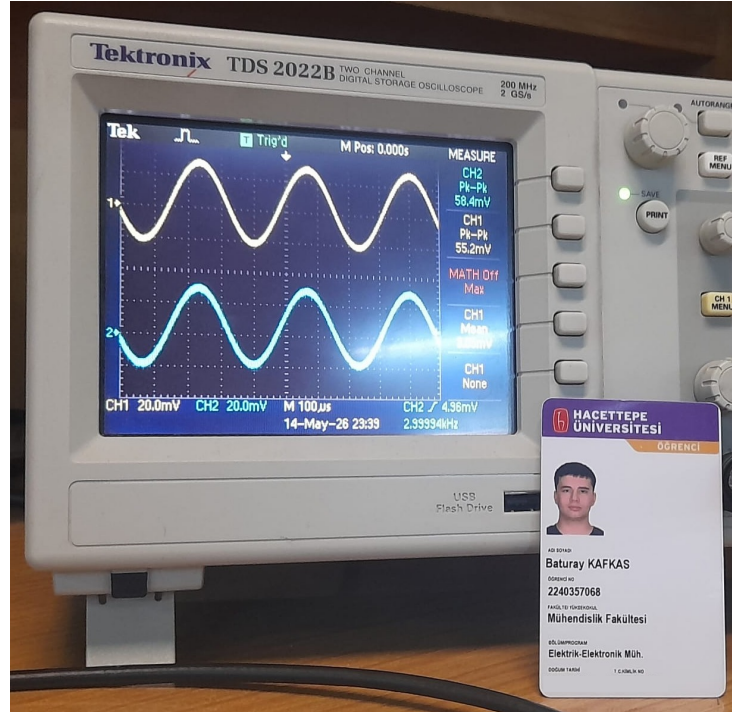


Figure 5.2: Output Voltage of the Buffer Stage on the Breadboard

For V_{in} set to 50 mV_{pp}, the input and output voltages are measured to be 55.2 mV and 58.4 mV, respectively, indicating a healthy buffer stage with a negligible difference.

Figure 5.3 shows the assembly of the gain stage on the breadboard and Figure 5.4 and Figure 5.5 demonstrate whether the gain stage works properly.

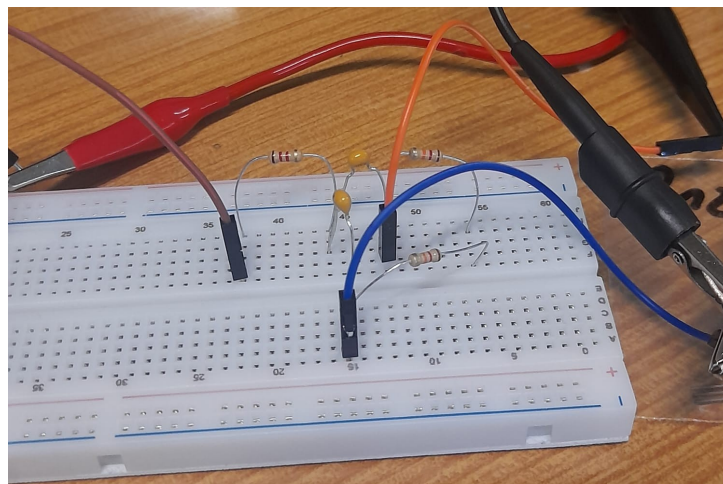


Figure 5.3: Equalizer Stage on the Breadboard

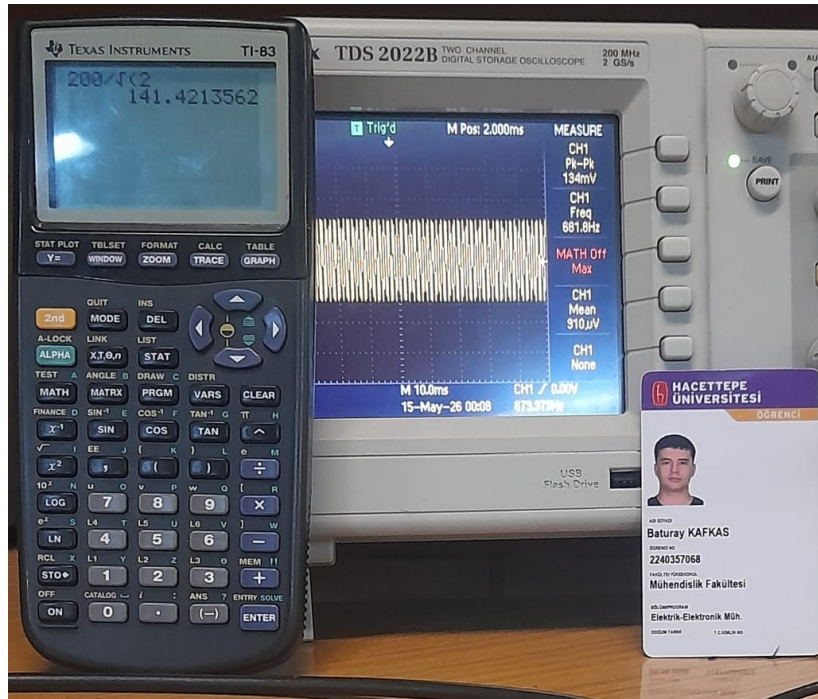


Figure 5.4: Equalizer Stage on the Breadboard Adjusted for Lower Cutoff Frequency

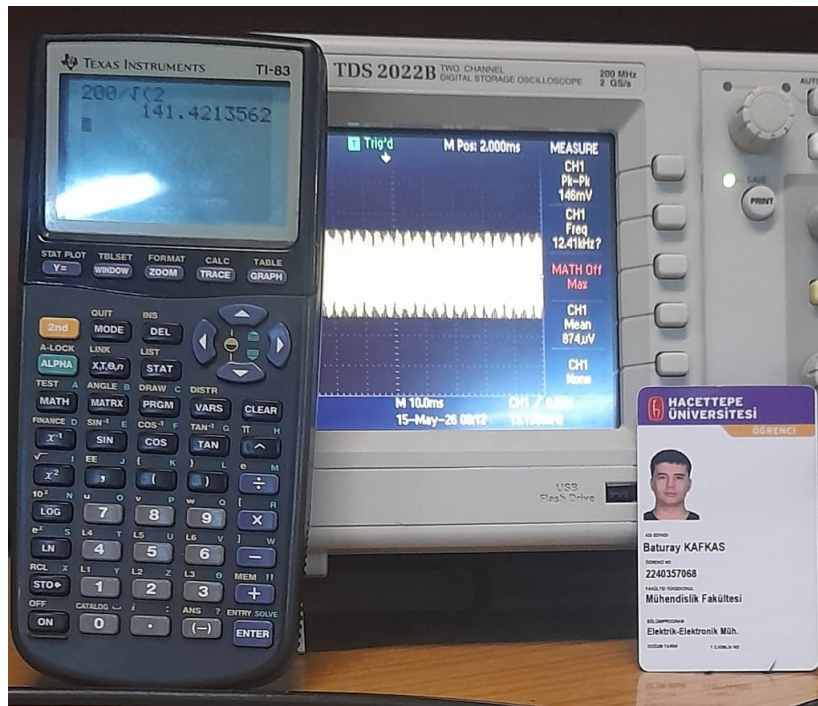


Figure 5.5: Equalizer Stage on the Breadboard Adjusted for Upper Cutoff Frequency

As seen above, the equalizer stage also works fine. The half of the power is consumed when these frequencies are applied.

Figure 5.6 shows the assembly of the gain stage on the breadboard and Figure 5.7 demonstrates whether the gain stage works properly. Figure 5.8 shows the assembly of the multistage amplifier on the breadboard and the results.

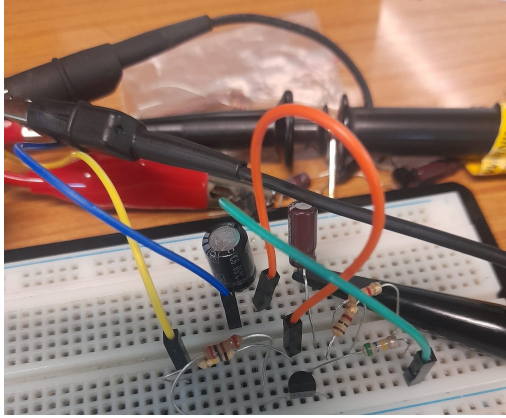


Figure 5.6: Gain Stage on the Breadboard

The gain obtained in this stage is

$$A_v = -408/54.4 \text{ V/V} = -7.5 \text{ V/V},$$

which is quite reasonable.

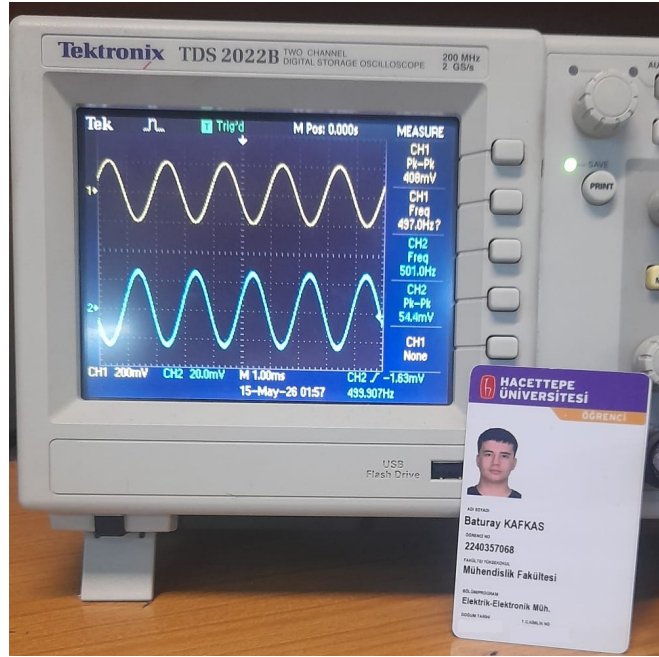


Figure 5.7: Output Voltage of the Gain Stage on the Breadboard

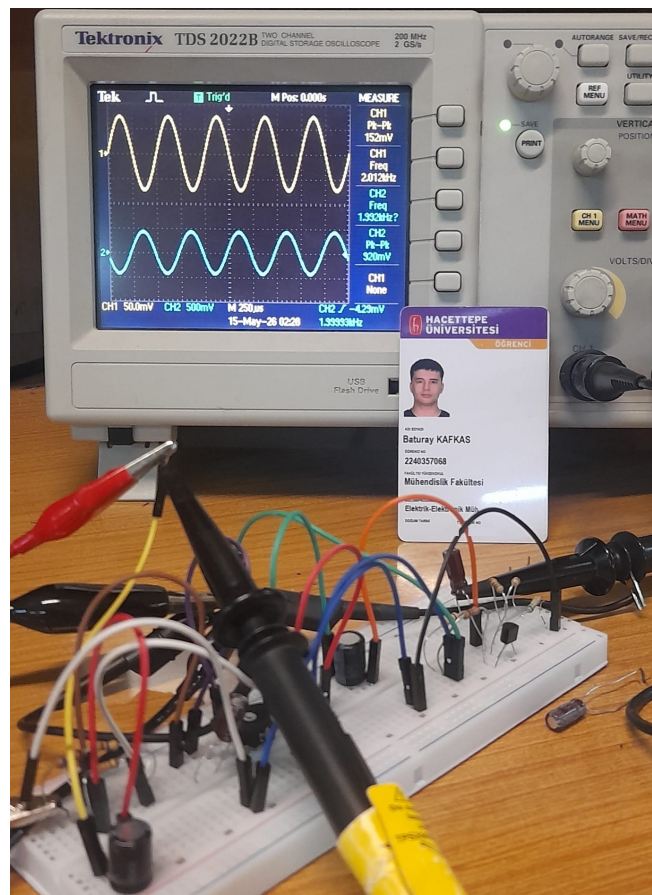


Figure 5.8: Multistage Audio Amplifier on the Breadboard and its Testing

The overall gain is measured to be $A_v = -920/152 \text{ V/V} \approx -6.05$, indicating a 25% loss.

Figure 5.9 and Figure 5.10 demonstrate the process of building the circuit on the breadboard.

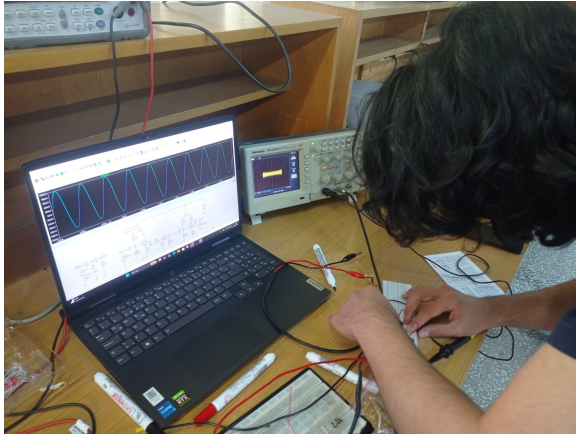


Figure 5.9: Breadboard Debugging

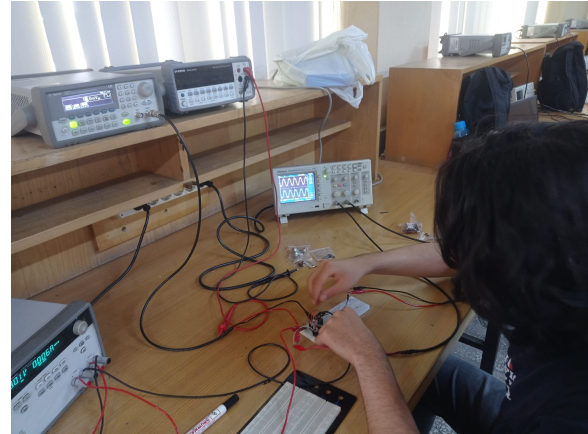


Figure 5.10: Breadboard Testing

5.2 Assembly on Pertinax

From the simulation and breadboard testing, we can conclude that the gain is most likely to be lower than the desired value. Therefore, we decided to add a series potentiometer to the drain line in order to adjust the gain.

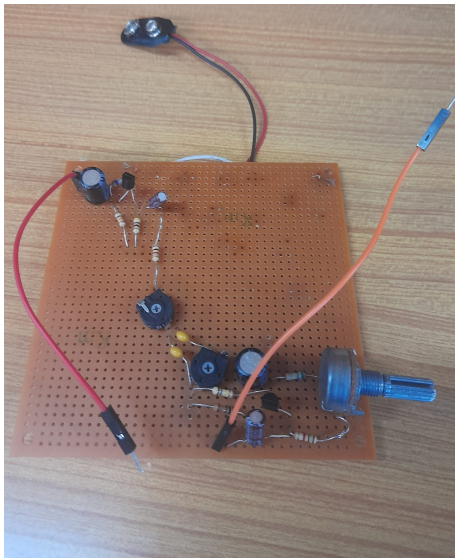


Figure 5.11: Circuit Fabricated on a Pertinax

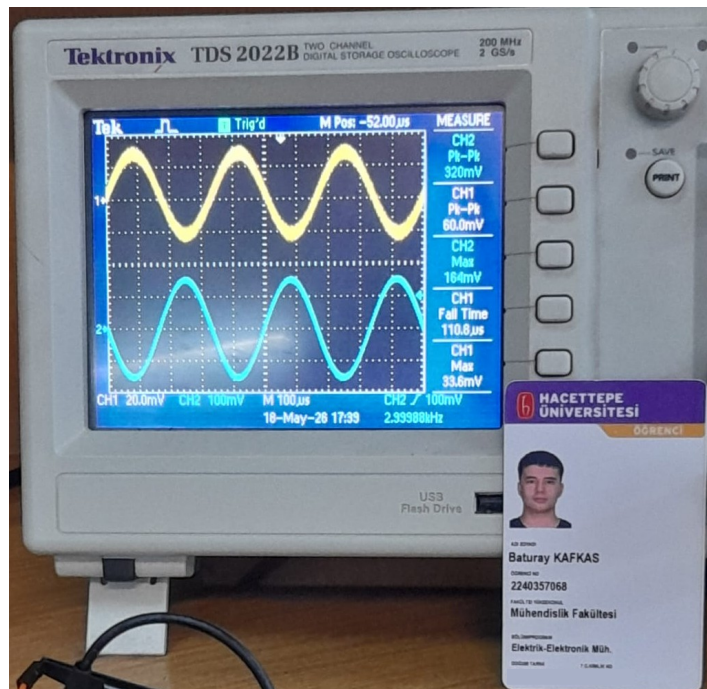


Figure 5.12: Testing of the Circuit Fabricated on a Pertinax

For an input voltage of 50 mV_{pp} , the gain is $A_v = -320/60.0 \text{ V/V} \approx -5.33 \text{ V/V}$. If the first channel had measured 50 mV_{pp} , we would obtain $A_v = -320/50 \text{ V/V} = -6.4 \text{ V/V}$.

Figure 5.13 and Figure 5.14 show the minimum and maximum gain that can be obtained with the introduction of the potentiometer, respectively.

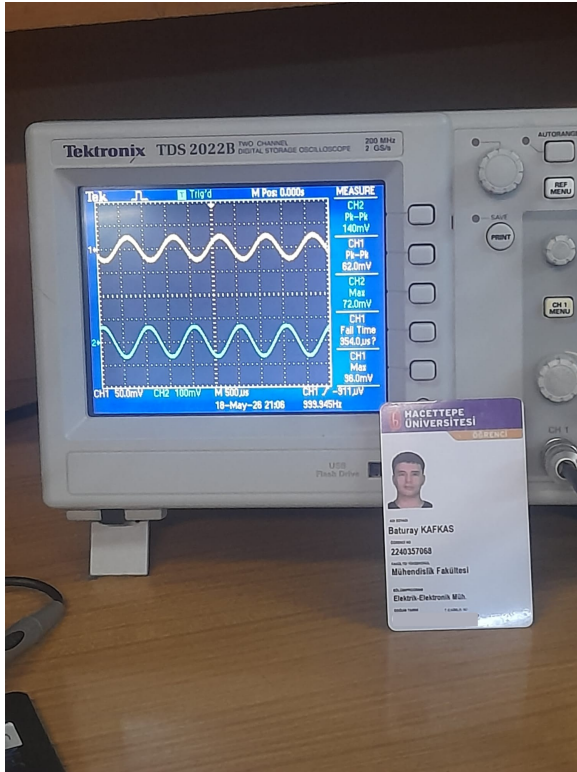


Figure 5.13: Circuit Fabricated on a Pertinax Adjusted for Minimum Gain

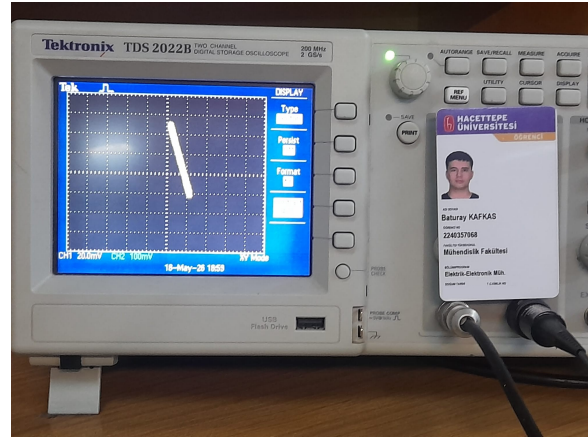


Figure 5.14: Circuit Fabricated on a Pertinax Adjusted for Maximum Gain

We can obtain a minimum gain of

$$|A_{v, \min}| = 140/62.0 \text{ V/V} \approx 2.26 \text{ V/V}$$

and a maximum gain of roughly

$$|A_{v, \max}| \approx 350/20.0 \text{ V/V} = 17.5 \text{ V/V}.$$

Figure 5.15 demonstrates the soldering process.



Figure 5.15: Soldering Process

6 Conclusion

In this project, we managed to design, simulate and fabricate a multi-stage audio amplifier incorporating a BJT buffer stage, a passive band-pass equalizer stage, and a MOSFET gain stage without the use of operational amplifiers. The design combined semiconductor principles with practical circuit considerations to meet specific targets governed by the student ID parameters ($f_{low} = 680$ Hz, $f_{high} = 13200$ Hz, and a gain of 8).

The selection of a 2N2222 BJT in an emitter-follower configuration effectively fulfilled the requirements of the buffer stage. The passive equalizer stage utilized linear and stable capacitive networks to form low- and high-pass filters, mitigating nonlinearities at the cost of minor signal attenuation ($\approx 9.8\%$). Finally, a 2N7000 NMOS transistor in a common-source configuration was employed, taking advantage of its exceptionally high gate isolation.

LTspice simulations confirmed the stable behavior of each individual block, yielding an equalizer bandpass close to theoretical values ($f_{lower} \approx 706$ Hz and $f_{upper} \approx 12.460$ kHz). While the individual common-source configuration provided a peak localized gain of $A_v \approx -8.38$ V/V, cascade loading interactions within the complete multistage layout resulted in a final simulated voltage gain of $A_v \approx -7.27$ V/V accompanied by the predicted 180° phase inversion.

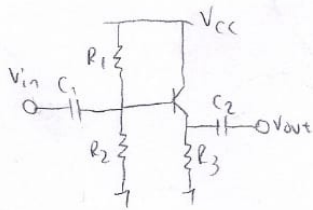
The most significant part of the assembly of the circuit on a Pertinax board was to apparently introduce a trimpot in series with the drain resistor to adjust the gain. The discrepancies between the analytical calculations, SPICE simulations, and final physical measurements can be attributed to factors such as internal component tolerances, parasitic contact resistances, and variations in transistor parameters. Eventually, the project successfully demonstrated the practical realization of discrete audio circuits.

7 References

- [1] D. A. Neamen, "Basic BJT Amplifiers," in *Design Application: Audio Amplifier*, 4th ed. New York, NY, USA: McGraw-Hill, 2010, pp. 445–449.
- [2] "2N2222 PDF," ALLDATASHEET.COM. [Online]. Available: https://www.alldatasheet.com/datasheet-pdf/view/888490/SEMTECH_ELEC/2N2222.html. [Accessed: 16 May, 2026].
- [3] onsemi, "NDS7002A - N-Channel Enhancement Mode Field Effect Transistor," datasheet, 1998. [Online]. Available: <https://www.onsemi.com/download/data-sheet/pdf/nds7002a-d.pdf>. [Accessed: May 16, 2026].
- [4] "Standard.mos," LTwiki. [Online]. Available: <https://ltwiki.org/index.php?title=Standard.mos>. [Accessed: May 16, 2026].

A Handwritten Calculations and Schematics

Buffer Stage Design



Assumptions: $I_C = 10 \text{ mA}$, $\beta = 200$, $V_{BE,ON} = 0.7 \text{ V}$, $I_{R2} \approx I_{R1} \geq 10 I_B$

$$I_B = \frac{I_C}{\beta} = \frac{10 \text{ mA}}{200} = 50 \mu\text{A}, \quad I_E = I_C + I_B \approx 10 \text{ mA}$$

$$\text{Take } I_{R1} = 10 I_B \Rightarrow I_{R1} = 0.5 \text{ mA}$$

$$R_1 + R_2 = \frac{V_{CC}}{I_{R1}} = \frac{9}{0.5 \text{ mA}} = 18 \text{ k}\Omega$$

$$\text{Choose } R_1 = 6.8 \text{ k}\Omega, \quad R_2 = 10 \text{ k}\Omega \Rightarrow R_1 + R_2 = 16.8 \text{ k}\Omega \approx 18 \text{ k}\Omega \text{ (standard value)}$$

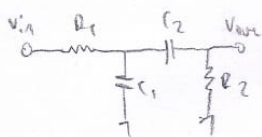
Emitter-Follower Configuration

$$V_E = V_B - V_{BE} = V_{CC} \cdot \frac{R_2}{R_1 + R_2} - V_{BE} = 9 \cdot \frac{10 \text{ k}\Omega}{16.8 \text{ k}\Omega} - 0.7 \approx 4.66 \text{ V}$$

$$V_E = I_E R_E \Rightarrow R_E = \frac{V_E}{I_E} \approx \frac{V_E}{I_C} \approx \frac{4.66 \text{ V}}{10 \text{ mA}} \approx 466 \Omega \Rightarrow R_E = 470 \Omega \text{ (standard value)}$$

Pick $C_1 = 1 \mu\text{F}$ and $C_2 = 10 \mu\text{F}$.

Equalizer Stage Design



$f_{\text{lower}} = 680 \text{ Hz}$, $f_{\text{upper}} = 13200 \text{ Hz}$

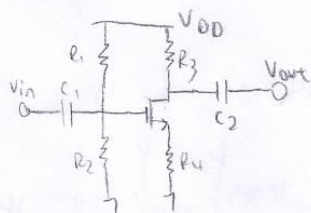
Let $C_1 = C_2 = 10 \text{ nF}$.

$$R_2 = \frac{1}{2\pi \cdot f_{\text{lower}} \cdot C_2} = \frac{1}{2\pi \cdot 680 \cdot 10 \cdot 10^{-9}} = 23.405 \text{ k}\Omega$$

$$\text{Choose } R_1 = 24 \text{ k}\Omega, \quad R_2 = 1.2 \text{ k}\Omega \text{ (standard values)}$$

$$R_1 = \frac{1}{2\pi \cdot f_{\text{upper}} \cdot C_1} = \frac{1}{2\pi \cdot 13200 \cdot 10 \cdot 10^{-9}} = 1.205 \text{ k}\Omega$$

Gain Stage Design



Assumptions: $K = 0.085 \text{ A/V}^2$, $V_{DS} = 2 \text{ V}$, $V_{TH} = 1.6 \text{ V}$, $I_{D,SAT} = 2 \text{ mA}$

Pick $C_1 = 1 \mu\text{F}$ and $C_2 = 10 \mu\text{F}$.

We assume the transistor in saturation for amplification. Therefore,

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2 = K (V_{GS} - V_{TH})^2$$

$$\Rightarrow 2 \text{ mA} = 0.085 (V_{GS} - 1.6)^2 \Rightarrow V_{GS} = 1.75 \text{ V}$$

$$g_m = 2K (V_{GS} - V_{TH}) = 2 \times 0.085 \times 0.15 = 26 \text{ mS}$$

Common-Source Configuration

$$|A_v| = 8 \Rightarrow \frac{g_m R_3}{1 + g_m R_4} = \frac{R_3}{1 + R_4} = 8 \Rightarrow R_3 = 8R_4 + 308 \quad (1)$$

DS loop: $V_{DD} = V_{DS} + I_D (R_3 + R_4)$

$$\Rightarrow R_3 + R_4 = \frac{V_{DD} - V_{DS}}{I_D} = \frac{9 - 4.5}{2 \text{ mA}} = 2250 \quad (2)$$

$$(1) \text{ and } (2) \text{ yields } R_3 = 2250 \Omega, \quad R_4 = 215 \Omega$$

$$\text{Choose } R_3 = 2.2 \text{ k}\Omega, \quad R_4 = 220 \Omega \text{ (standard value)}$$

$$\text{GS loop: } V_{DD} \cdot \frac{R_2}{R_1 + R_2} - V_{GS} - I_D R_4 = 0 \Rightarrow \frac{1}{\frac{R_1}{R_2} + 1} = \frac{I_D R_4 + V_{GS}}{V_{DD}} = \frac{2 \times 10^{-3} \times 220 + 1.75}{9}$$

$$\Rightarrow \frac{R_1}{R_2} = \frac{227}{73} \approx 3.11 \quad \text{Take } R_1 = 560 \text{ k}\Omega, \quad R_2 = 180 \text{ k}\Omega \Rightarrow \frac{R_1}{R_2} = \frac{560 \text{ k}\Omega}{180 \text{ k}\Omega} \approx 3.11 \text{ (standard values)}$$

Figure A.1: Handwritten Analytical Calculations

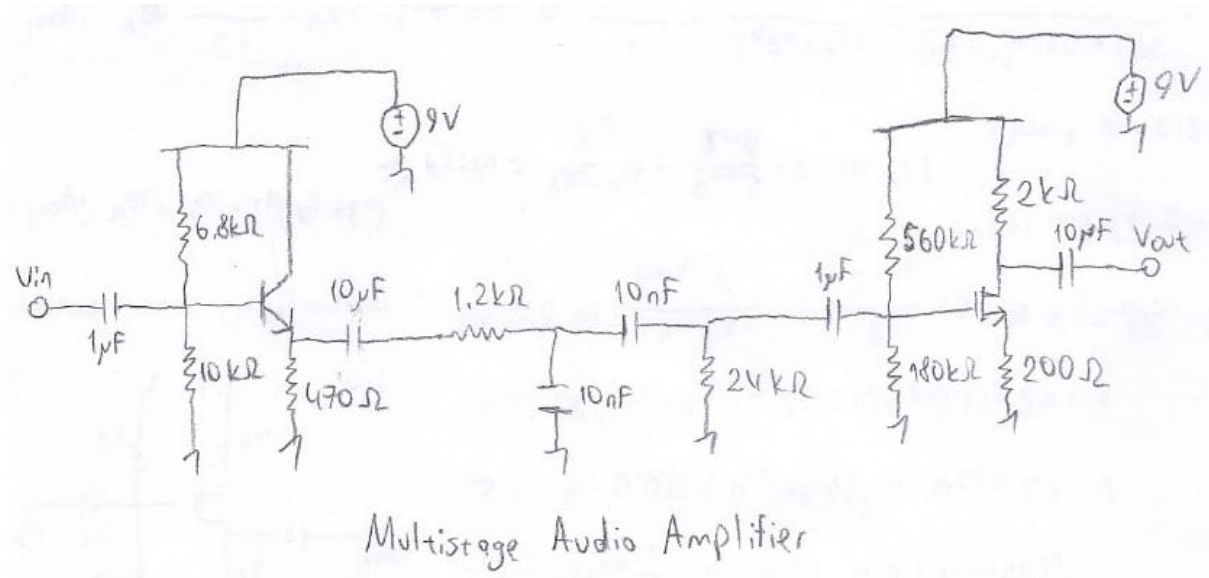


Figure A.2: Handwritten Multistage Audio Amplifier

B Figures and Tables

Table B.1: Timestamps for Fabrication and Testing Figures

Figure	Description	Timestamp
5.1	Buffer Stage on the Breadboard	14/05/2026 18:23
5.2	Output Voltage of the Buffer Stage on the Breadboard	14/05/2026 18:23
5.3	Equalizer Stage on the Breadboard	14/05/2026 18:51
5.4	Equalizer Stage on the Breadboard Adjusted for Lower Cutoff Frequency	14/05/2026 18:51
5.5	Equalizer Stage on the Breadboard Adjusted for Upper Cutoff Frequency	14/05/2026 18:55
5.6	Gain Stage on the Breadboard	14/05/2026 20:40
5.7	Output Voltage of the Gain Stage on the Breadboard	14/05/2026 20:40
5.8	Multistage Audio Amplifier on the Breadboard and its Testing	14/05/2026 21:12
5.9	Breadboard Debugging	May 2026
5.10	Breadboard Testing	May 2026
5.11	Circuit Fabricated on a Pertinax	18/05/2026 14:50
5.12	Testing of the Circuit Fabricated on a Pertinax	18/05/2026 11:16
5.13	Circuit Fabricated on a Pertinax Adjusted for Minimum Gain	18/05/2026 14:43
5.14	Circuit Fabricated on a Pertinax Adjusted for Maximum Gain	18/05/2026 12:36
5.15	Soldering Process	May 2026

List of Tables

3.1	Comparison between BJT and MOSFET	4
3.2	Electrical Characteristics of 2N2222/2N2222A NPN Transistor ($T_a = 25^\circ C$ unless otherwise noted)	5
3.3	Electrical Characteristics of 2N7000 MOSFET ($T_C = 25^\circ C$ unless otherwise noted)	6
3.4	Buffer Stage Component Values	8
3.5	Equalizer Stage Component Values	8
3.6	Gain Stage Component Values	8
B.1	Timestamps for Fabrication and Testing Figures	21

List of Figures

3.1	LTspice Symbols From Left to Right: NPN, PNP, NMOS, PMOS	4
3.2	Beta Values for Different Temperature Values	7
3.3	Relationship between the Collector Current and the Base-Emitter Voltage	7
4.1	Buffer Stage Simulation Circuit	9
4.2	Buffer Stage Simulation	10
4.3	Equalizer Stage Simulation Circuit	10
4.4	Equalizer Stage Simulation (Voltage Characteristics)	11
4.5	Equalizer Stage Simulation (Voltage Characteristics) Comparison	11
4.6	Equalizer Stage Simulation (Bode Plot)	11
4.7	Equalizer Stage Simulation (Bode Plot) Result	11
4.8	Gain Stage Simulation Circuit	12
4.9	Gain Stage Simulation	12
4.10	Gain Stage Simulation Result	12
4.11	Multistage Simulation Circuit	13
4.12	Multistage Simulation	13
4.13	Multistage Simulation Result	13
5.1	Buffer Stage on the Breadboard	14
5.2	Output Voltage of the Buffer Stage on the Breadboard	14
5.3	Equalizer Stage on the Breadboard	14
5.4	Equalizer Stage on the Breadboard Adjusted for Lower Cutoff Frequency	15
5.5	Equalizer Stage on the Breadboard Adjusted for Upper Cutoff Frequency	15
5.6	Gain Stage on the Breadboard	16
5.7	Output Voltage of the Gain Stage on the Breadboard	16
5.8	Multistage Audio Amplifier on the Breadboard and its Testing	16
5.9	Breadboard Debugging	17
5.10	Breadboard Testing	17
5.11	Circuit Fabricated on a Pertinax	17
5.12	Testing of the Circuit Fabricated on a Pertinax	17
5.13	Circuit Fabricated on a Pertinax Adjusted for Minimum Gain	18
5.14	Circuit Fabricated on a Pertinax Adjusted for Maximum Gain	18
5.15	Soldering Process	18
A.1	Handwritten Analytical Calculations	20
A.2	Handwritten Multistage Audio Amplifier	21